GAUTHAM SAI

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LinkedIn | GitHub | Portfolio

Profile

Electronics and Communication Engineering student focused on **embedded systems and hardware security**, with hands-on experience in PCB design, firmware development, embedded security, and automotive security. Experienced in taking embedded hardware from design and prototyping through firmware integration and testing. Hardware Team Lead at **Team bi0s**, with practical security research and technical speaking experience at **Black Hat India and IndiaFOSS**.

Education

B.Tech. Electronics and Communication Engineering — *2024 – Present*
Amrita Vishwa Vidyapeetham, Amritapuri — **CGPA: 8.89**

Higher Secondary Education — *2021 – 2023*
Sacred Heart Higher Secondary School, Dwaraka

Projects

Vanguard — INCTF 2026 Hardware Security Badge

- Led the hardware and firmware development of the **Vanguard security badge**, owning custom PCB design, embedded firmware, hardware integration, and deployment.
- Engineered an **ESP32-S3-WROOM-1** platform with **16 MB flash, NVS-backed persistence, LoRa, and BLE** for embedded functionality and security-focused applications.
- Built a complete **four-level onboard hardware CTF** and implemented security challenges for the INCTF platform.
- Developed build, flashing, monitoring, provisioning, and deployment workflows for repeatable badge setup.

CAN Bus Communication & Security Toolkit

- Built a **three-node CAN testbed** using Arduino Nano and MCP2515/TJA1050 modules to simulate ECU communication on a **500 kbps** automotive network.
- Created a dedicated security tool supporting **CAN traffic sniffing, message injection, and denial-of-service flooding**, including high-priority frame transmission to demonstrate CAN arbitration abuse.
- Demonstrated disruption of legitimate ECU communication by manipulating CAN traffic and exploiting arbitration-based bus access.

RF Fingerprinting for IoT Device Authentication

- Built an RF fingerprinting framework for **same-model Wi-Fi IoT device authentication**, using device-specific characteristics extracted from raw I/Q signals.
- Trained and evaluated Random Forest and CNN classifiers across **30 devices and ∼30,000 samples**, achieving **99.89% and 89.71% accuracy**, respectively, on clean data.
- Evaluated robustness across **0–20 dB SNR**; Random Forest retained **89.04% accuracy at 20 dB** without retraining.
- Conducted label-permutation tests for spoofed device identities, with both models dropping to **0.74% accuracy**.

PeerDrop — BLE-Based Peer-to-Peer System

- Built a BLE peer-to-peer data exchange system using **RSSI-based proximity validation** and dynamic session authentication.
- Integrated ESP32 NVS-based identity storage and one-time transfer validation to prevent unauthorized and repeated exchanges.

Smart Anti-Tremor Glove

- Designed and prototyped a wearable embedded system using an **MPU6050 IMU, five vibration motors, and Bluetooth** for real-time tremor detection and suppression.
- Implemented threshold-based detection with a **0.08 g** activation threshold and PWM-based vibration feedback for tremor-like oscillations in the **4–7 Hz** range.
- Integrated Bluetooth telemetry at **20 Hz** with **<80 ms delay** during a 15-minute continuous test without data loss; published at **IEEE IDCIoT 2026**.

Skills

Core Security: Hardware Security, Embedded Security, Side-Channel Analysis, Automotive Security, CTF Development
Hardware Design: PCB Design, Schematic Design, Circuit Design, Hardware Bring-up, Debugging & Troubleshooting
Embedded Systems: ESP32, ESP-IDF, STM32, Arduino, Raspberry Pi, LPC2148, Firmware Development
Protocols: CAN, UDS, UART, SPI, I2C, BLE, Wi-Fi, LoRa, MQTT
Programming: C, Embedded C, Python
Tools: KiCad, ChipWhisperer, Wireshark, GNU Radio, PulseView/Sigrok, LTspice, MATLAB, Linux, Git, GitHub
Non-Technical: Technical Leadership, Team Coordination, Mentoring, Public Speaking, Technical Presentation, Project Planning, Research, Technical Documentation

Experience

Hardware Team Lead — Team bi0s *2025 – Present*

- Lead a **hardware team** working across embedded security, hardware security, automotive security, firmware, and embedded development.
- Drive security-focused hardware projects from system architecture and PCB design through firmware integration, hardware bring-up, testing, and deployment.
- Mentor team members through hands-on embedded and hardware security projects, research, and practical security exercises.
- Coordinate technical development and contribute to security platforms, CTF infrastructure, and demonstrations across embedded and hardware security domains.

Publications

An IoT-based Smart Glove for Real-Time Tremor Detection and Suppression in Parkinson's Patients
Gautham Sai, Hrishikesh S., Gunisha Kaur, Gowrinandana R. Kaimal, Aswathy K. Nair, Prajitha S.
Published at the **IEEE International Conference on Distributed Computing and Internet Technology (IEEE IDCIoT 2026)**.

Technical Speaking

Arsenal Speaker — Black Hat India 2026

- Presented **"Sysrupt: Portable OT Cyber Range"**, a hands-on platform for exploring operational technology and industrial control security.

Technical Speaker — IndiaFOSS 2026, Security Devroom

- Delivered **"Extracting AES Keys from Embedded Devices Using Open Source Hardware"**, demonstrating practical side-channel analysis and AES key extraction from embedded devices.

Technical Speaker — bi0s Meetup

- Conducted an offline session, **"Introduction to Side-Channel Analysis"**, combining fundamentals with a practical demonstration of physical leakage from embedded devices.

Achievements

- **2nd Runner-Up — Caterpillar Tech Challenge 2026**, National Finals; contributed to the development of an LSTM-based adaptive PI controller using MATLAB/Simulink.
- **1st Place — TWIST Hack 2025**, FOSS Fest.
- **1st Prize — IGNITE 2025 Ideathon**, Amrita Vishwa Vidyapeetham.

Certifications

ISEA–ISAP 2026 — Security & Privacy Conference, IIT Madras
Embedded Systems with STM32 — Anokha 2024
Semiconductor Workshop — Vidyut 2025

Languages

English | Malayalam | Tamil | Hindi